Canonical Optimal Wallets Have at Most One Incomplete Layer

1 Model and statement

Let $[n] = \{1, \dots, n\}$. Independently for each key, let its state be

state	owner has the key	attacker has the key	probability
safe	1	0	s
stolen	0	1	t
leaked	1	1	e
lost	0	0	ℓ

where

$$s + t + e + \ell = 1, \qquad s > \frac{1}{2}.$$

Write $O, D \subseteq [n]$ for the sets of keys held by the owner and the attacker, respectively. A *wallet* is an upward-closed family $\mathcal{F} \subseteq 2^{[n]}$. Its success probability is

$$\Phi_{s,t,e,\ell}(\mathcal{F}) = \mathbb{P}(O \in \mathcal{F},\ D \notin \mathcal{F}).$$

When the parameters are fixed, we abbreviate this to $\Phi(\mathcal{F})$.

For $0 \le k \le n$, let

$$\mathcal{F}_k = \mathcal{F} \cap \binom{[n]}{k}.$$

The kth layer is called *incomplete* if

$$0 < |\mathcal{F}_k| < \binom{n}{k}.$$

Theorem 1. *For every wallet $\mathcal{F} \subseteq 2^{[n]}$, there is a wallet $\mathcal{F}^\star \subseteq 2^{[n]}$ such that*

$$\Phi(\mathcal{F}^\star) \ge \Phi(\mathcal{F})$$

and $\mathcal{F}^\star$ has at most one incomplete layer. Consequently, some optimal wallet has at most one incomplete layer.

The proof uses three batch edits at the two extreme incomplete layers. The main ingredient is a rank-kernel inequality that remains valid in the presence of leaked keys.

2 The three batch edits

Suppose that $\mathcal{F}$ has at least two incomplete layers, and let $r < q$ be the lowest and highest incomplete ranks. Upward closure implies that every layer below r is empty and every layer above q is full. Put

$$\mathcal{A} = \mathcal{F}_r, \qquad \mathcal{B} = \binom{[n]}{q} \setminus \mathcal{F}_q,$$

and define

$$\alpha = \frac{|\mathcal{A}|}{\binom{n}{r}}, \qquad \beta = \frac{|\mathcal{B}|}{\binom{n}{q}}.$$

Both α and β are strictly between zero and one. Consider the three wallets

$$\mathcal{F}_R = \mathcal{F} \setminus \mathcal{A}, \qquad \mathcal{F}_Q = \mathcal{F} \cup \mathcal{B}, \qquad \mathcal{F}_{RQ} = (\mathcal{F} \setminus \mathcal{A}) \cup \mathcal{B}.$$

They are upward closed. Indeed, removing the lowest nonempty layer cannot violate upward closure, and filling the highest non-full layer cannot violate it.

Let

$$I = |O|, \qquad J = |D|,$$

and, for integers k, m, define

$$g_k(m) = \mathbb{P}(I \geq m, J = k) - \mathbb{P}(I = k, J < m). \tag{1}$$

Lemma 2 (Batch-edit bounds)**.** *The changes caused by the three edits satisfy*

$$\Phi(\mathcal{F}_R) - \Phi(\mathcal{F}) \geq \alpha g_r(q+1), \tag{2}$$
$$\Phi(\mathcal{F}_Q) - \Phi(\mathcal{F}) \geq -\beta g_q(r), \tag{3}$$
$$\Phi(\mathcal{F}_{RQ}) - \Phi(\mathcal{F}) \geq L_1 := \alpha g_r(q) - \beta g_q(r), \tag{4}$$
$$\Phi(\mathcal{F}_{RQ}) - \Phi(\mathcal{F}) \geq L_2 := \alpha g_r(q+1) - \beta g_q(r+1). \tag{5}$$

Proof. Conditioned on $I = k$, the owner set O is uniform on $\binom{[n]}{k}$; conditioned on $J = k$, the attacker set D is uniform on $\binom{[n]}{k}$. The same statements remain true after imposing any condition involving only (I, J). This follows directly from the exchangeability of the n coordinates.

For the remove edit, a success is certainly gained whenever $D \in \mathcal{A}$ and $I \geq q+1$, because all layers above q are full. A success can be lost only when $O \in \mathcal{A}$ and $J \leq q$, because an attacker with at least $q+1$ keys was already authorized. Uniformity within a rank therefore gives

$$\Phi(\mathcal{F}_R) - \Phi(\mathcal{F}) \geq \alpha\big[\mathbb{P}(I \geq q+1, J = r) - \mathbb{P}(I = r, J < q+1)\big],$$

which is (2).

For the fill edit, a success is certainly gained whenever $O \in \mathcal{B}$ and $J < r$, because every layer below r is empty. A success can be lost only when $D \in \mathcal{B}$ and $I \geq r$, because every authorized owner has at least r keys. Thus

$$\Phi(\mathcal{F}_Q) - \Phi(\mathcal{F}) \geq \beta\big[\mathbb{P}(I = q, J < r) - \mathbb{P}(I \geq r, J = q)\big],$$

which is (3).

There are two ways to perform the joint edit. If layer q is filled first, the fill step contributes at least $-\beta g_q(r)$. In the resulting wallet, layer q is full, so the subsequent removal of $\mathcal{A}$ contributes at least $\alpha g_r(q)$. This proves (4). If $\mathcal{A}$ is removed first, the removal step contributes at least $\alpha g_r(q+1)$. In the resulting wallet every layer through rank r is empty, so subsequently filling $\mathcal{B}$ contributes at least $-\beta g_q(r+1)$. This proves (5). □

The elementary discrete derivative

$$g_k(m+1) - g_k(m) = -\mathbb{P}(I = m, J = k) - \mathbb{P}(I = k, J = m) \leq 0 \tag{6}$$

will be used repeatedly.

3 The low-sum kernel certificate

Proposition 3 (Kernel certificate)**.** *If*

$$1 \le r < q \le n-1, \qquad r+q \le n,$$

then

$$\boxed{\binom{n}{q} g_r(q+1) \ge \binom{n}{r} g_q(r+1).} \tag{7}$$

We first show that this certificate implies the desired batch improvement when $r+q \le n$.

Lemma 4 (Low-sum batch improvement)**.** *If $r+q \le n$, at least one of $\mathcal{F}_R, \mathcal{F}_Q, \mathcal{F}_{RQ}$ has success probability at least $\Phi(\mathcal{F})$.*

Proof. If $g_r(q+1) \ge 0$, (2) shows that the remove edit is non-worsening. If $g_q(r) \le 0$, (3) shows that the fill edit is non-worsening. It remains to consider

$$g_r(q+1) < 0 < g_q(r). \tag{8}$$

Set

$$w = \mathbb{P}(I = q, J = r) + \mathbb{P}(I = r, J = q)$$

and

$$\rho = \frac{\binom{n}{q}}{\binom{n}{r}}.$$

Since $r < q \le n-r$, binomial unimodality and symmetry give $\rho \ge 1$. By (6),

$$g_r(q+1) = g_r(q) - w, \qquad g_q(r+1) = g_q(r) - w. \tag{9}$$

The certificate (7) says

$$\rho\big(g_r(q) - w\big) \ge g_q(r) - w.$$

Rearranging,

$$g_r(q) - g_q(r) \ge (\rho - 1)\big(w - g_r(q)\big).$$

The first inequality in (8) and (9) imply $w > g_r(q)$, so

$$g_r(q) \ge g_q(r) > 0. \tag{10}$$

Using (9) again gives

$$g_q(r+1) \le g_r(q+1) < 0. \tag{11}$$

If $L_1 \ge 0$, (4) proves that the joint edit is non-worsening. Otherwise

$$\alpha g_r(q) < \beta g_q(r).$$

By (10), this implies $\alpha < \beta$. By (11),

$$-g_q(r+1) \ge -g_r(q+1) > 0.$$

Consequently

$$\begin{aligned} L_2 &= \beta[-g_q(r+1)] - \alpha[-g_r(q+1)] \\ &\ge (\beta - \alpha)[-g_r(q+1)] > 0. \end{aligned}$$

Now (5) proves that the joint edit is improving. □

It remains to prove Proposition 3. We do this coefficientwise.

4 Coefficient reduction

Put

$$\eta = s - (t + e + \ell) = 2s - 1 \geq 0, \qquad s = t + e + \ell + \eta. \tag{12}$$

After this substitution, the difference between the left- and right-hand sides of (7) is a homogeneous polynomial in t, e, ℓ, η. We shall prove that every coefficient is nonnegative.

Fix a monomial

$$t^a e^b \ell^c \eta^d, \qquad a + b + c + d = n,$$

and fix a labelled partition

$$[n] = T \sqcup E \sqcup L \sqcup H, \qquad |T| = a, \quad |E| = b, \quad |L| = c, \quad |H| = d.$$

In the expansion (12), this partition has the following interpretation:

block	allowed states
T	safe or stolen
E	safe or leaked
L	safe or lost
H	safe only.

Call a choice of one of the allowed states at every coordinate a *toggle assignment*. If x, y, z denote the numbers chosen as stolen, leaked, and lost in the first three blocks, respectively, then

$$J = x + y, \qquad n - I = x + z, \tag{13}$$

and the multiplicity of these choices is

$$\binom{a}{x}\binom{b}{y}\binom{c}{z}.$$

For this fixed labelled partition define

$$G_k(m) = \#\{J = k,\ n - I \leq n - m\} - \#\{n - I = n - k,\ J < m\}, \tag{14}$$

where both counts range over all toggle assignments. The coefficient of $t^a e^b \ell^c \eta^d$ in the polynomial from (7) is the common positive multinomial factor

$$\frac{n!}{a!\, b!\, c!\, d!}$$

times

$$\binom{n}{q} G_r(q+1) - \binom{n}{r} G_q(r+1). \tag{15}$$

Thus Proposition 3 follows once (15) is shown to be nonnegative for every labelled partition.

5 Averaging over nested sets

For $A \subset B \subseteq [n]$, with $|A| = r$ and $|B| = q$, define

$$\begin{aligned}\Lambda(A,B) = \mathbf{1}_{\{D=A,\ |O|\geq q+1\}} + \mathbf{1}_{\{O=B,\ |D|\leq r\}} \\ - \mathbf{1}_{\{O=A,\ |D|\leq q\}} - \mathbf{1}_{\{D=B,\ |O|\geq r+1\}}.\end{aligned} \tag{16}$$

Summing over all toggle assignments and all nested pairs gives

$$\begin{aligned}&\sum_{\substack{A\subset B\\ |A|=r,\ |B|=q}} \sum_{\text{toggle assignments}} \Lambda(A,B) \\ &\quad = \binom{n-r}{q-r} G_r(q+1) - \binom{q}{r} G_q(r+1) \\ &\quad = \frac{\binom{q}{r}}{\binom{n}{r}} \left[\binom{n}{q} G_r(q+1) - \binom{n}{r} G_q(r+1) \right].\end{aligned} \tag{17}$$

Indeed, a fixed r-set occurs as the lower member of $\binom{n-r}{q-r}$ nested pairs, while a fixed q-set contains $\binom{q}{r}$ lower r-sets; the identity in the last line follows by double-counting pairs (A, B) with $A \subset B$.

We prove that the left-hand side of (17) is nonnegative. Fix

$$M = B \setminus A, \qquad |M| = q - r,$$

and sum over all r-sets A disjoint from M. There are four exhaustive types.

1. If M contains an H-coordinate, both negative events in (16) are impossible: an H-coordinate belongs to O and does not belong to D in every toggle assignment.

2. If M contains both an E-coordinate and an L-coordinate, then $O = A$ is impossible because every E-coordinate belongs to O, whereas $D = B$ is impossible because no L-coordinate belongs to D.

3. If every coordinate of M lies in T, switching all coordinates of M from stolen to safe gives injections

   $$\{O = A,\ |D| \leq q\} \longrightarrow \{O = B,\ |D| \leq r\}$$

   and

   $$\{D = B,\ |O| \geq r+1\} \longrightarrow \{D = A,\ |O| \geq q+1\}.$$

   The inverse on either image switches the coordinates of M back from safe to stolen, so both maps are injective.

4. The only remaining types are

   $$M \subseteq T \cup L, \quad M \cap L \neq \varnothing,$$

   and

   $$M \subseteq T \cup E, \quad M \cap E \neq \varnothing.$$

The next two sections handle these final two types.

6 The L-case

Assume

$$M \subseteq T \cup L, \qquad l := |M \cap L| \geq 1.$$

Outside M, let the numbers of T, E, L, H coordinates be

$$T_0, \quad b, \quad L_0, \quad d,$$

respectively. Put

$$\delta = n - r - q \geq 0, \qquad k_0 = r - b - d.$$

Since $|M| = q - r$, elementary counting gives

$$T_0 = 2r - b - L_0 - d + \delta, \qquad 2k_0 = T_0 - b + L_0 - d - \delta. \tag{18}$$

We first record the event count that will be used. In the negative event $O = A$, every outside-M coordinate of $E \cup H$ is forced into A. If A contains j of the outside L-coordinates, it must contain $k_0 - j$ outside T-coordinates. Pairing this event with the positive event $O = B$ leaves precisely the configurations in which the number of leaked outside E-coordinates is

$$L_0 - \delta - j + s, \qquad 1 \leq s \leq l.$$

For the other positive event, $D = A$, reindex by

$$|A \cap E| = L_0 - \delta - j.$$

The required T-choice has size complementary to $k_0 - j$, and therefore has the same binomial count $\binom{T_0}{k_0 - j}$. If W is the set of lost coordinates in the combined L-block of size $L_0 + l$, the condition $|O| \geq q + 1$ is exactly

$$|W| \leq L_0 - j - 1.$$

Consequently, after summing (16) over all A disjoint from this fixed M, the desired nonnegativity is equivalent to

$$\sum_{j \in \mathbb{Z}} \binom{T_0}{k_0 - j} H_j \geq 0, \tag{19}$$

where

$$\begin{aligned} H_j = &\binom{b}{L_0 - \delta - j} \sum_{w=0}^{L_0 - j - 1} \binom{L_0 + l}{w} \\ &- \binom{L_0}{j} \sum_{s=1}^{l} \binom{b}{L_0 - \delta - j + s}. \end{aligned} \tag{20}$$

As usual, an out-of-range binomial coefficient and an empty sum are zero. In particular, j ranges over all integers.

We prove (19) by an injection from the objects counted by the negative part to those counted by the positive part. A negative object indexed by (j, s) consists of

$$\begin{aligned} K \subseteq [T_0], \quad & |K| = k_0 - j =: k, \\ J \subseteq [L_0], \quad & |J| = j, \\ A_E \subseteq [b], \quad & |A_E| = L_0 - \delta - j + s, \\ & 1 \leq s \leq l. \end{aligned}$$

Set

$$j' = j - s.$$

Keep A_E, and let

$$W = ([L_0] \setminus J) \cup P_{s-1}, \tag{21}$$

where P_{s-1} is the canonical prefix of length $s-1$ in the fixed, ordered l-element set $M \cap L$. Then

$$|W| = L_0 - j + s - 1 = L_0 - j' - 1,$$

so W is counted by the cumulative sum in the positive term of (20) at index j'.

It remains to raise K from size k to size $k+s$. Nonvanishing of the negative object implies

$$L_0 - \delta - j + s \le b.$$

Using (18), we obtain

$$\begin{aligned} T_0 - 2k - s &= b - L_0 + d + \delta + 2j - s \\ &= (b - L_0 + \delta + j - s) + (d + j) \ge 0. \end{aligned} \tag{22}$$

Hence $k + s \le T_0 - k$. Fix a symmetric-chain decomposition of $2^{[T_0]}$. Moving upward by s positions in the chain containing K defines an injective map

$$\binom{[T_0]}{k} \longrightarrow \binom{[T_0]}{k+s};$$

condition (22) guarantees that the target position exists. Let K' be the image of K.

The resulting triple (K', A_E, W) is a positive object at index j'. It recovers s from the length of the canonical prefix $W \cap (M \cap L)$, then recovers j, J, and A_E, and finally recovers K from the injective raising map. Thus the map is injective, and images belonging to different (j, s) are disjoint. This proves (19).

7 The E-case

Assume

$$M \subseteq T \cup E, \qquad e_M := |M \cap E| \ge 1.$$

Outside M, let the numbers of T, E, L, H coordinates be

$$T_0, \quad E_0, \quad c, \quad d,$$

respectively, and put

$$\delta = n - r - q, \qquad k_0 = r - E_0 - d.$$

We again first record the event count. In the positive event $O = B$, suppose A contains j outside L-coordinates. It must contain every outside E- and H-coordinate, leaving $k_0 - j$ choices from the outside T-block. The condition $|D| \le r$ permits at most

$$c - \delta - j$$

leaked coordinates in the combined E-block of size $E_0 + e_M$.

For the negative event $D = B$, first pair all configurations covered by the positive event $D = A$ by forcing the coordinates of M safe. Reindex the remaining configurations by

$$|A \cap E| = c - \delta - j.$$

The unmatched configurations have an outside-L safe set of size

$$j + 1 - s, \qquad 1 \le s \le e_M.$$

It follows that the fixed-M sum is

$$\sum_{j\in\mathbb{Z}} \binom{T_0}{k_0 - j} \left[\binom{c}{j} \sum_{w=0}^{c-\delta-j} \binom{E_0 + e_M}{w} - \binom{E_0}{c - \delta - j} \sum_{s=1}^{e_M} \binom{c}{j + 1 - s} \right]. \tag{23}$$

A negative object indexed by (j, s) consists of

$$\begin{aligned} K \subseteq [T_0], \quad & |K| = k_0 - j =: k, \\ A_E \subseteq [E_0], \quad & |A_E| = c - \delta - j, \\ J \subseteq [c], \quad & |J| = j + 1 - s, \\ & 1 \le s \le e_M. \end{aligned}$$

Set

$$j' = j + 1 - s.$$

Keep J, extend A_E by the canonical prefix of length $s - 1$ in the ordered set $M \cap E$, and raise K by $s - 1$ ranks in the fixed symmetric-chain decomposition of $2^{[T_0]}$.

The required raising map exists. Indeed, nonvanishing of A_E and J gives

$$E_0 - c + \delta + j \ge 0, \qquad j - s + 1 \ge 0.$$

The analogue of (18) therefore yields

$$T_0 - 2k - (s - 1) = (E_0 - c + \delta + j) + (d + j - s + 1) \ge 0. \tag{24}$$

Thus $k + s - 1 \le T_0 - k$, as required. The enlarged E-set has size

$$c - \delta - j + s - 1 = c - \delta - j',$$

which is the top allowed size in the positive cumulative term of (23) at index j'.

The canonical prefix records s. Hence the image recovers j, A_E, and J, and injectivity of the symmetric-chain raising map recovers K. Images from different (j, s) are disjoint. Therefore the negative objects in (23) inject into the positive objects, so (23) is nonnegative.

Every fixed-M contribution in (17) is now nonnegative. It follows that (15) is nonnegative for every labelled partition. Every coefficient of the polynomial obtained after (12) is therefore nonnegative. Since $t, e, \ell, \eta \ge 0$, the polynomial itself is nonnegative. This proves Proposition 3.

8 The high-sum case by duality

Lemma 5 (High-sum batch improvement)**.** *If $r + q > n$, at least one of $\mathcal{F}_R, \mathcal{F}_Q, \mathcal{F}_{RQ}$ has success probability at least $\Phi(\mathcal{F})$.*

Proof. Define the dual wallet

$$\mathcal{F}^* = \{X \subseteq [n] : [n] \setminus X \notin \mathcal{F}\}.$$

It is upward closed. Define transformed owner and attacker sets by

$$O^* = [n] \setminus D, \qquad D^* = [n] \setminus O.$$

Coordinatewise, this transformation preserves safe and stolen states and exchanges leaked and lost states. Moreover,

$$O^* \in \mathcal{F}^*,\ D^* \notin \mathcal{F}^* \quad \Longleftrightarrow \quad O \in \mathcal{F},\ D \notin \mathcal{F}.$$

Consequently

$$\Phi_{s,t,e,\ell}(\mathcal{F}) = \Phi_{s,t,\ell,e}(\mathcal{F}^*). \tag{25}$$

The lowest and highest incomplete ranks of $\mathcal{F}^*$ are

$$r^* = n - q, \qquad q^* = n - r.$$

Since $r + q > n$,

$$r^* + q^* = 2n - r - q < n.$$

Lemma 4, applied to $\mathcal{F}^*$ with the leaked and lost probabilities interchanged, gives a non-worsening remove, fill, or joint edit of $\mathcal{F}^*$. Under duality, removal at the lowest incomplete layer becomes filling at the highest incomplete layer of $\mathcal{F}$, filling becomes removal, and the joint edit remains joint. Equation (25) therefore gives a non-worsening batch edit of $\mathcal{F}$. □

9 Completion of the proof

Proof of Theorem 1. If $\mathcal{F}$ already has at most one incomplete layer, take $\mathcal{F}^\star = \mathcal{F}$. Otherwise, let $r < q$ be its extreme incomplete ranks. Lemma 4 or Lemma 5, according as $r + q \le n$ or $r + q > n$, supplies one of the three batch edits whose success probability is at least $\Phi(\mathcal{F})$.

The remove edit makes layer r empty, the fill edit makes layer q full, and the joint edit does both. No other layer changes. Hence each batch edit eliminates at least one incomplete layer and creates none. After finitely many steps, the process terminates at a wallet $\mathcal{F}^\star$ with at most one incomplete layer and

$$\Phi(\mathcal{F}^\star) \ge \Phi(\mathcal{F}).$$

Finally, begin with an optimal wallet. Every non-worsening edit must then preserve the optimal value. The terminal wallet is therefore optimal and has at most one incomplete layer. □